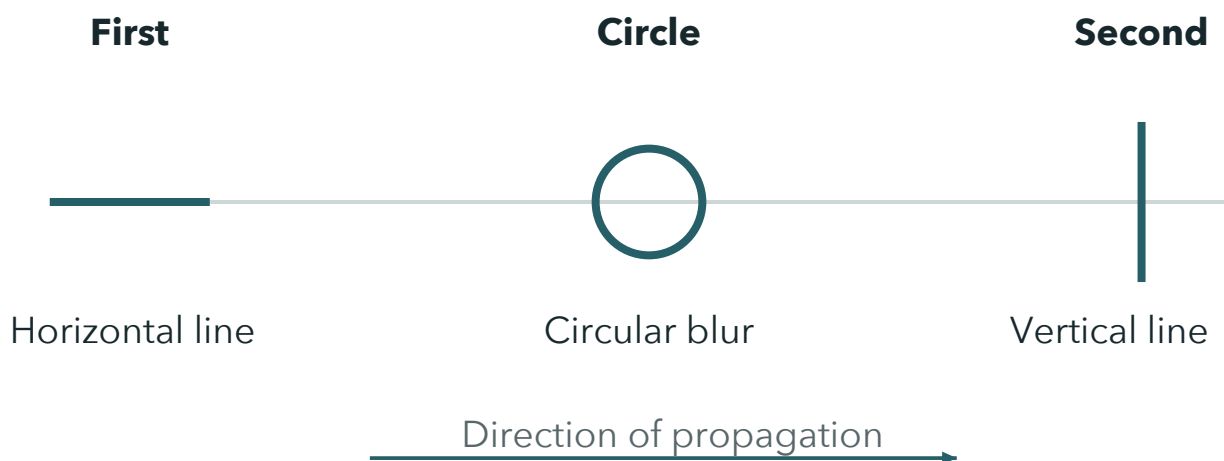


The conoid of Sturm

Part 1 | A meridian focuses; the residual image is a line

The first focal line is perpendicular to the stronger focusing meridian.



For stronger vertical power, the first line is horizontal; the second is vertical, with the circular section between.

Keep two orientations separate

The meridian names the direction in which rays are bent. At its focus, that direction's spread collapses while the perpendicular spread remains. This explains why focal-line orientation is not simply the axis label of the stronger meridian.

Focal order follows power

For collimated input in air, greater positive power gives the nearer meridional focus.

An original example

Meridian	Power	Focal distance
Vertical	+10 D	10 cm
Horizontal	+5 D	20 cm

Predict before calculating

The first focal line is horizontal.

The second focal line is vertical.

Between them, the blur passes from horizontal ellipse through a circular section to vertical ellipse.

The circular section is not a common point focus

Equal blur dimensions can remain nonzero.

Plane

Feature	Image section	What remains
First focal line	Line	Perpendicular blur
Least-confusion plane	Circle	Residual astigmatic blur
Second focal line	Orthogonal line	Other meridional blur

Interpretation

Moving the observation plane can change blur orientation without changing the lens's cylinder. A circular blur therefore does not prove that the system is free of astigmatism. For a paraxial model, work in vergence when locating the least-confusion plane; do not simply average focal distances.

Cylinder correction collapses the interval

A sphere shifts both meridians; a cylinder changes their difference.

Reason through correction

- Identify the principal powers
- ↓
- Add equal sphere to both
- ↓
- Add cylinder in its powered meridian
- ↓
- Recalculate the residual meridional difference

Original comparison

A +2 D spherical change added to principal powers +6 and +8 D gives +8 and +10 D: their difference remains 2 D. A correctly oriented -2 D cylinder applied to the stronger meridian can equalise them. The two adjustments answer different optical problems.

Regular and irregular astigmatism differ

A two-meridian conoid is a useful model with explicit limits.

Model

Feature	Assumption	Limitation
Regular astigmatism	Two orthogonal principal meridians	Cylinder representation is useful
Irregular optics	Spatially varying wavefront	Two powers do not capture every defect
Higher-order terms	Beyond sphere/cylinder	May persist after refraction

Examination approach

Use the regular model when its assumptions are supplied. If the question introduces spatially irregular corneal optics, do not force every distortion into two focal lines; distinguish an imperfect model from an arithmetic error within a valid model.